GATE CS & IT Previous 12-Year Chapter-wise Solved Question Bank

High-weightage concept capsules and rigorous step-by-step solutions covering Data Structures, Operating Systems, DBMS, Theory of Computation, and Computer Networks.

LITHAVEN CAMPUS STUDY VAULT • OFFICIAL REVISION CAPSULE
EXAMINATION: GATE Computer Science & Information Technology | **WEIGHTAGE:** 100 Marks Benchmark Syllabus

1. Subject Weightage & Core High-Yield Scoring Areas

Discipline / Subject	Average Marks	Highest Frequency Core Topics	Recommended Strategy
Data Structures & Algorithms	14 - 16 Marks	Asymptotic Analysis, Trees, Heaps, Graph traversals, DP.	Solve all Master Theorem and recursion tree problems.
Operating Systems	8 - 10 Marks	CPU Scheduling, Semaphores/Peterson's, Paging address translation.	Master virtual memory page size & inverted table calculations.
DBMS	7 - 9 Marks	Relational Algebra, SQL subqueries, Normalization, Serializable schedules.	Practice conflict serializability precedence graphs.
Theory of Computation (TOC)	8 - 10 Marks	DFA state minimization, Closure properties, Decidability, CFG.	Memorize Chomsky hierarchy closure property matrix.
Computer Networks	7 - 9 Marks	Subnetting / CIDR, TCP Congestion Control, Sliding Window Flow Control.	Learn Go-Back-N and Selective Repeat efficiency formulas.

2. Selected Benchmark Solved GATE Problems

Problem 1: Master Theorem Recurrence (GATE CS)

Question: Solve T(n) = 3T(n/2) + Theta(n^2). Find the tight asymptotic bound.
Solution: Here a = 3, b = 2, f(n) = n^2. Compute log_b(a) = log_2(3) approx 1.585.
Since f(n) = Omega(n^{1.585 + eps}) for eps = 0.415, and 3(n/2)^2 = (3/4)n^2 <= c*n^2 (regularity holds), by Master Theorem Case 3: **T(n) = Theta(n^2)**.

Problem 2: Two-Level Paging Address Translation (GATE CS)

Question: 32-bit virtual address, 4 KB page size, 4 bytes per page table entry. Find size of second-level page table.
Solution: Page offset = log_2(4 KB) = 12 bits. Virtual page number = 32 - 12 = 20 bits.
Entries per page = 4 KB / 4 B = 1024 = 2^10 entries (10 bits for page table).
Level 1 page index = 10 bits, Level 2 page index = 10 bits. Second-level page table size = 2^10 * 4 B = **4 KB**.

Problem 3: Sliding Window Protocol Maximum Efficiency (GATE CS)

Question: Packet size = 1 KB, Bandwidth = 1.5 Mbps, Round Trip Time (RTT) = 45 ms. Find minimum window size W for 100% link utilization.
Solution: Transmission time T_t = (1024 * 8 bits) / (1.5 * 10^6 bps) = 5.46 ms. Propagation delay T_p = RTT / 2 = 22.5 ms.
Parameter a = T_p / T_t = 22.5 / 5.46 = 4.12. Minimum window size W = 1 + 2a = 1 + 2(4.12) = 9.24. Round up to **10 frames**.